EXAMINING EMOTION MALLEABILITY BELIEFS AS A MODERATOR ON ASSOCIATIONS BETWEEN EARLY LIFE MALTREATMENT AND POOR SLEEP

THE OHIO STATE UNIVERSITY | THE AFFECTIVE SCIENCE LAB

Keegan Whitacre, Mallory Feldman, Adrienne Bonar, Ruofan Ma, & Kristen A. Lindquist

SOCIETY FOR AFFECTIVE SCIENCE

INTRODUCTION

- **Maltreatment → poor sleep** across the lifespan (Brown et al., 2022; Abajobir et al., 2017)
- **Emotion regulation** disrupted by early adversity, linked to sleep (Gruhn & Compas, 2020)
- **Emotion beliefs** shape regulation strategy use and outcomes (Kim et al., 2024)
- **Interoception** as candidate mechanism linking adversity to biased perception (Ditzer et al., 2025)
- **Aim 1:** do emotion beliefs buffer maltreatment sleep link?
- **Aim 2:** does maltreatment affect perceived vs objective sleep differently?

METHODS

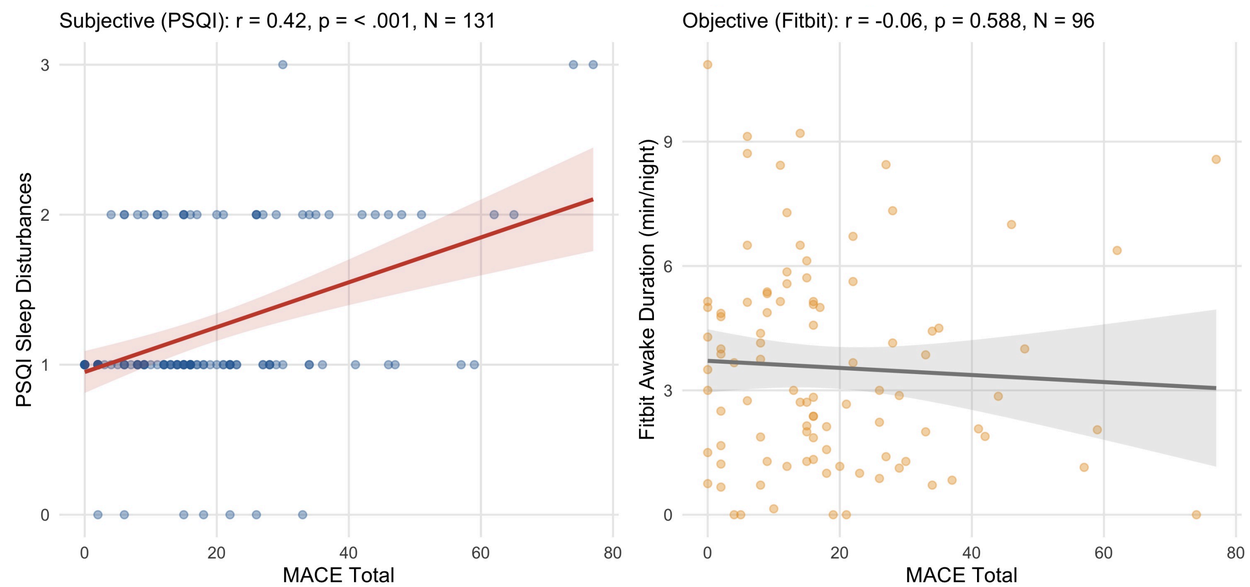
- **Sample:** N = 165 adults (ages 18–83); Fitbit data (n = 96); heartbeat discrimination (HBD) task (n = 70) after quality filter
- **Maltreatment:** MACE (10 subscales + total score)
- **Sleep:** PSQI (subjective) + Fitbit ~7 nights (objective: awake duration, efficiency, fragmentation)
- **Emotion beliefs:** Personal Beliefs about Emotion Malleability (PBEM) scale
- **Interoception:** Heartbeat discrimination task, d' sensitivity index (signal detection framework)
- **Discrepancy score:** z(PSQI disturbances) − z(Fitbit awake duration); positive = perceives worse than objective
- **Analysis:** OLS regression, HC3 robust SEs, FDR correction (BH) for subscale sweeps; ordered logit sensitivity checks

RESULTS

NULL FINDING: EMOTION MALLEABILITY BELIEFS

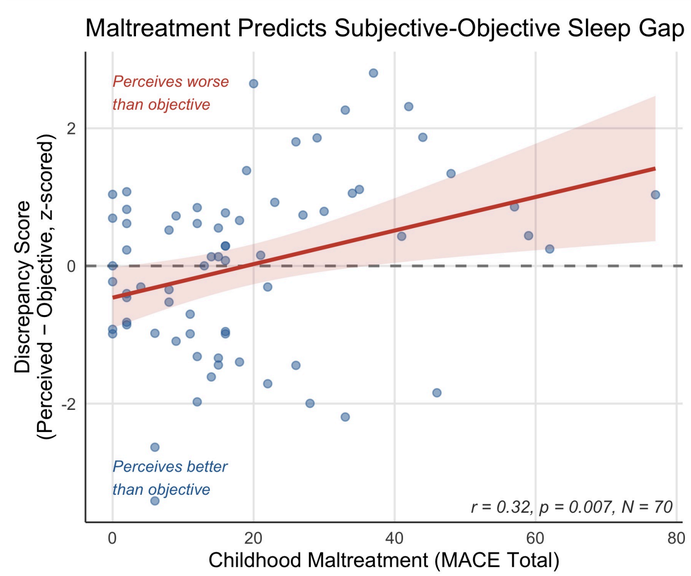
PBEM did not moderate or mediate any maltreatment → sleep association (all interaction $p > .50$; ACME ≈ 0, 95% BCI spans zero). Maltreatment did not predict PBEM ($\beta = -.005$, $p = .281$). Simple slopes show the MACE effect on disturbances is significant across the entire range of emotion beliefs — beliefs **do not** buffer this pathway at any level.

MALTREATMENT PREDICTS PERCEIVED — NOT OBJECTIVE — SLEEP DISRUPTION

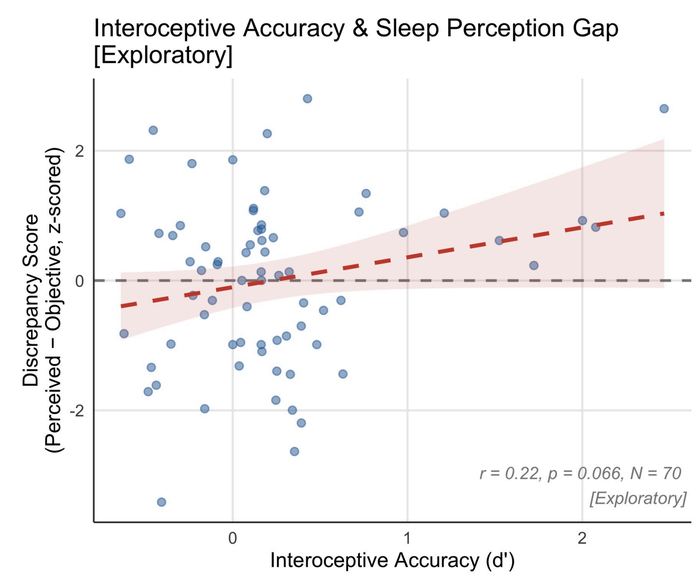


Greater maltreatment predicts a larger subjective-objective sleep gap ($r = .32$, $p = .007$, $N = 70$; $R^2 = .23$). Age independently predicts a smaller gap ($\beta = -.029$, $p = .002$) — older adults perceive more accurately despite worse objective sleep; the MACE effect on the gap is attenuated at older ages (interaction $p = .126$).

MALTREATMENT → PERCEPTION GAP



EXPLORATORY: INTEROCEPTIVE ACCURACY



DISCUSSION

- **Maltreatment → perceived but not objective sleep** — suggests a subjective-objective gap rather than uniformly worse sleep
- **Theory of Constructed Emotion** (Barrett, 2017): the brain constructs affective experiences from interoceptive predictions shaped by prior experience (Feldman, Bliss-Moreau & Lindquist, 2024); we speculate that early adversity may bias these predictions toward threat, contributing to a subjective-objective sleep gap
- **Age → objective fragmentation** (not perceived) — consistent with interoceptive recalibration across lifespan
- **Higher interoceptive accuracy → larger perception gap:** may reflect sensitivity to genuine physiological states that accelerometer-based wearables do not capture (cf. Bynum & Brindle, 2025) ; interoceptive sensitivity has downstream clinical relevance in adversity-exposed populations (Reid-Russell & Nock, 2024)

FUTURE DIRECTIONS

- **Replicate** subjective-objective dissociation with larger N and pre-registration
- **Formal mediation:** MACE → interoception → perception gap
- **Compare** objective interoceptive accuracy (d') vs. subjective sensibility (BAQ) as predictors
- **Daily diary:** night-to-night variability in sleep perception

CONTACT